ORESTE AFFATATO

Researcher in Medicine and Scientific Methods

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I work on understanding the reasons why we get sick.

EDUCATION

Ph.D. degree in Medicine - Neuroscience

Uppsala University

June 2024 Uppsala, Sweden

Thesis: "Beating of hammers"
I investigated the pathophysiology of migraine and depression, using many approaches, from physiology to occupational epidemiology. During my doctoral studies, I read mostly courses in neuroscience, mental health, epidemiology and statistics

M.Sc. in Physics

University of Milan

April 2020 Milan, Italy

Thesis: "Survival of early replicators under transient compartmentalization"
I specialized in theoretical physics, mathematical physics and mathematical modelling applied to biology

B.Sc. in Physics

University of Milan-Bicocca

July 2017 Milan, Italy

Thesis: "Nonlinear field equations: symmetries and solitons"
I read standard Bachelor courses in physics and mathematics. I took elective courses in theoretical physics, mathematical physics and numerical analysis

RESEARCH EXPERIENCE

Postdoctoral Researcher

University of Gothenburg

March 2025 – Ongoing Gothenburg, Sweden

Ph.D. studies

Uppsala University

September 2020 – September 2024 Uppsala, Sweden

TEACHING EXPERIENCE

Program in Biomedicine

Uppsala University

2021 - 2023

- Statistics, mathematical modelling and basic R programming

EXPERTISE

Neuroscience | Neurology | Psychiatry | Psychology | Epidemiology | Causal Inference | Bayesian Methods | Statistics | Applied Mathematics

STRENGHTS

Programming (R, Python, Matlab, LaTeX) | Mathematical modelling | Statistical Analysis | Study Design | Microsoft Office

Project Management | Problem Solving | Active Listening | Critical Thinking | Teamwork | Independece | Motivation | Positive Attitude | Public Speaking | Research Collaboration | Self Learning | Knowledge Sharing

LANGUAGES

"Alle volte uno si crede incompleto ed è soltanto giovane." - Italo Calvino

"I measured out my life with coffee spoons." - Thomas Stearns Eliot

"Eu tinha a grande saúde de não perceber coisa nenhuma." - Álvaro de Campos

"Los espejos y la cópula son abominables porque multiplican el número de los hombres." - Jorge Luis Borges

"Döden är vindstilla." - Thomas Tranströmer

- Co-supervision of Bachelor's thesis projects

Medical Program

Uppsala University

2021 - 2023

- Journal clubs, seminars on psychiatry and neurology

PUBLICATIONS

Journal Articles

- M. Gentreau, M. Miguet, **O. Affatato**, G. Rukh, and H. B. Schiöth, "Statin use is associated with higher white matter hyperintensity volumes and lower grey matter volumes," *Brain Communications*, vol. 6, 2024. DOI: http://dx.doi.org/10.1093/braincomms/fcae417.
- S. L. Muataz, J. Mwinyi, **O. Affatato**, *et al.*, "Exploring sex differences: Insights into gene expression, neuroanatomy, neurochemistry, cognition, and pathology," *Frontiers in Neuroscience*, vol. 18, 2024. DOI: 10.3389/fnins.2024.1340108.
- **O. Affatato**, A. Dahlén, G. Rukh, H. Schiöth, and J. Mwinyi, "Assessing volumetric brain differences in migraine and depression patients: A uk biobank study," *BMC Neurology*, vol. 23, 2023. DOI: https://doi.org/10.1186/s12883-023-03336-x.
- **O. Affatato**, G. Rukh, H. Schiöth, and J. Mwinyi, "Volumetric differences in cerebellum and brainstem in patients with migraine: A uk biobank study," *Biomedicines*, vol. 11, 2023. DOI: https://doi.org/10.3390/biomedicines11092528.
- **O. Affatato**, M. M, S. HB, and M. J., "Major sex differences in migraine prevalence among occupational categories: A cross-sectional study using uk biobank," *Journal of Headache and Pain*, vol. 4, 2021. DOI: 10.1186/s10194-021-01356-x.
- **O. Affatato**, M. TC, P. C, *et al.*, "High efficacy of onabotulinumtoxina treatment in patients with comorbid migraine and depression: A meta-analysis," *J Transl Med*, vol. 19, 2021. DOI: 10.1186/s12967-021-02801-w.
- F. Cabral, **O. Affatato**, M. de Oliveira, and C. Dias, "Modelagem matemática e computacional no crescimento do glioblastoma multiforme," *Brazilian Journal of Development*, vol. 6, 2020. DOI: 10.34117/bjdv6n3-096.

Books

- **O. Affatato**, *Beating of hammers*. Acta Universitatis Upsaliensis, 2024.

GRANTS AND AWARDS

- Foundation Blanceflor - Short-term scholarship (2026)
- Svenska Sällskapet för Medicinsk Forskning Postdoctoral Grant (2025)
- Sederholm, Nordic travel scholarships (2024)
- Busoli Gian Carlo and Birgit grant (2022)

RELEVANT EXPERIENCE

- **Guest Researcher at NTNU:** Research conducted as visiting PhD student in January 2024 at the Norwegian University of Science and Technology (NTNU) in Trondheim. Supervisors: prof. Leif Edward Ottesen Kennair and prof. Thomas Haarklau Kleppestø, department of Psychology (Sederholm, Nordic travel scholarships).

"Je suis l'Empire à la fin de la décadence" - Paul Verlaine

SCIENCE OUTREACH

Science Fika

Stadsbiblioteket i Uppsala

I am a co-founder of Science Fika. This association promotes science popularization through public talks at Uppsala (Sweden). We invite researchers with different backgrounds to give public talks to disseminate knowledge and passion for scientific inquiry.
Website: Science Fika

OPEN SCIENCE

More on my research projects and teaching materials (such as R scripts and supplementary information) can be found on my GitHub page: OresteAffatato

REFEREES

Available upon request

INTERESTS

Music | Theater | Film | Literature | Contemporary dance | Literary Criticism | Comedy | Art Exhibitions | Anthropology | Performance Art | Popular Science | And everything else

HOBBIES

Playing piano | Reading | Running | Swimming | Martial Arts | All the rest

- **European Educational Programme in Epidemiology:** 35th Residential 3-week Summer Course in Epidemiology 2022 in Fiesole (Italy) - biostatistics and epidemiology training (Busoli Gian Carlo and Birgit grant).
- **International Summer school in functional pharmacology 2022 - FAT4BRAIN:** Teacher in the workshop on statistics and data visualization with R - Latvian Institute of Organic Synthesis, Rīga (Latvia).
- **Erasmus studies:** One year (academic year 2018/2019) as an Erasmus+ Student at the University of Coimbra (Portugal).

ABOUT ME

A thirst for knowledge is the major driving force of my life. I delve into any subject with the utmost passion and drive, from the classics of world's literature, to the theories of space and time, and to the traits in our body that make us vulnerable to diseases. Ideas, perspectives and methods from different fields helped me immensely in framing research questions and in finding diverse ways to address them. Cultivating interest in a wide range of disciplines is for me a major resource for thinking outside the box, and growing as a scientist and as a person.
Moreover, I believe that knowledge has an immense transformative power and for this reason I am also a passionate educator. I put a great effort in my teaching activities both inside and outside the academic environment. My aim is to discuss ideas, challenge points of view, foster curiosity and ignite passion for scientific inquiry. Ultimately, I believe that our research, our knowledge, should be used to transform our society in a better version of itself. And better for everyone.